

AI-POWERED MEETING INTELLIGENCE

MeetingLens

Automated Summarization, Action Item Extraction & Task Delegation Platform

Transforming Spoken Conversations into Structured Knowledge & Verifiable Execution

Domain: Artificial Intelligence • Natural Language Processing • Full-Stack Web App

Problem Statement: The Meeting Productivity Crisis

1. Information Loss & Fatigue

- 30-60+ minute meetings overwhelm human working memory.
- Crucial context, decisions, and technical reasoning are lost immediately after calls end.
- Audio/video recordings are rarely rewatched due to lack of time.

2. Manual Note-Taking Flaws

- Designated note-takers cannot fully participate in active brainstorming.
- Human notes are subjective, prone to personal bias, and often incomplete.
- Inconsistent note formats across teams prevent centralized searchability.

3. Task Accountability Void

- >40% of assigned action items in meetings fall through the cracks.
- Verbal task assignments lack clear deadline, ownership, and priority definitions.
- No built-in verification mechanism to review and confirm tasks before delegation.

Expected Outcome & Proposed Solution: MeetingLens

Core Proposed Solution

Multi-Modal Meeting Capture: Record live audio with real-time Speech-to-Text or upload existing audio, video, and text logs.

Multi-Tier AI Summaries: Generate concise Executive Summaries, detailed chronological breakdowns, and key decision logs.

Automated Task Extraction: NLP engine detects tasks, assigns owners, identifies deadlines, and tags urgency levels automatically.

Key Measurable Deliverables

Human-in-the-Loop Task Approval: Interactive verification modal allowing managers to edit, confirm, or reassign AI tasks before saving.

Meeting Health & Quality Scoring: Objective metrics evaluating meeting efficiency, structure, and outcome clarity (0-100%).

Centralized Knowledge Archive: Instant searchable repository with downloadable reports (PDF, Markdown, JSON).

End-to-End Workflow Diagram

Step 1

Audio Ingestion

Live Voice Capture or
Audio / Video / Text
Upload

Step 2

Speech-to-Text

Real-time client
transcription & text
preprocessing

Step 3

AI Analysis Core

LLM extracts summaries,
action items & quality
score

Step 4

Task Approval

Human-in-the-loop review,
edit & owner assignment

Step 5

Persistence & Sync

Save to MySQL DB,
dispatch alerts & export
reports

Workflow Data Flow Highlights

Microphone Stream / File -> Speech Engine -> Clean Transcript -> Gemini / LLM Prompt -> Structured JSON -> Task Approval Modal -> MySQL Database -> Action Dashboard & Export Hub

System Architecture Diagram

1. Client Layer (React SPA)

- Live Voice Meeting Room (Web Speech API)
- Meeting Details & Dynamic Transcripts
- Task Approval & Delegation Modal
- Analytics & Executive Reports Dashboard

2. API & Gateway Layer (PHP REST)

- JWT Authentication & Session Guard
- /api/meetings - CRUD & History Engine
- /api/analyze - AI Prompt Orchestration
- /api/action-items - Task State & Delegation

3. Intelligence Layer (AI / LLM)

- Google Gemini / LLM NLP Core
- Structured JSON Schema Extraction
- Context Cleaning & Repair Engine
- Meeting Quality & Sentiment Scoring

4. Data Persistence (MySQL)

- users (Accounts, Credentials, Roles)
- meetings (Transcripts, Summaries, Scores)
- action_items (Tasks, Assignees, Deadlines)
- Relational Foreign Key Integrity (InnoDB)

Technologies & Tools Planned / Implemented

Technology Stack Breakdown

Layer	Technology / Tool	Justification / Role
Frontend UI	React.js (Vite), Tailwind / Vanilla CSS	High-speed rendering, modular reactive components & responsive design
Audio / STT	Web Speech API, MediaRecorder API	Zero-latency in-browser real-time speech capture & audio processing
Backend Server	PHP 8.x (RESTful API), Apache / XAMPP	Lightweight, robust API routing, JWT validation & prompt sanitization
Database	MySQL (InnoDB Relational Schema)	ACID-compliant storage for users, transcripts, summaries & action items
AI / NLP Core	Google Gemini API / Multi-Model LLM	Zero-shot JSON task extraction, executive summaries & quality scoring

Pseudocode: Meeting Intelligence Pipeline (MIP)

Algorithm Execution Flow

Algorithm: MeetingIntelligencePipeline(InputSource, CurrentUser)

1. IF InputSource is LiveAudio THEN Transcript = WebSpeechTranscribe(InputSource)
ELSE Transcript = SanitizeAndExtractText(InputSource)
2. CleanText = RemoveNoiseAndFillerWords(Transcript)
3. PromptPayload = BuildStructuredPrompt(Role='Executive PM', Transcript=CleanText, Schema='JSON')
4. AI_Response = InvokeLLM_API(PromptPayload)
5. ParsedData = ValidateAndParseJSON(AI_Response)
6. DB_Transaction_Begin():
MeetingID = InsertMeeting(user_id=CurrentUser.id, title=ParsedData.title, summary=ParsedData.summary,
executive_summary=ParsedData.executive_summary, quality_score=ParsedData.quality_score)
FOR EACH task IN ParsedData.action_items:
InsertActionItem(meeting_id=MeetingID, description=task.desc, assignee=task.owner, deadline=task.due)
DB_Transaction_Commit()
7. Trigger TaskApprovalModal(MeetingID, ActionItems) for Human Verification & Editing
8. Return Structured Meeting Dashboard View

Assumptions, Technical Challenges & Mitigations

Assumptions

Hardware & Network: Users have an active microphone input and internet connectivity for LLM API transactions.

Participant Identification: Speakers state names during discussion or context clues allow AI attribution.

Language Baseline: Primary input language is English, with support for standard business terminology.

Challenges & Mitigations

LLM Hallucinations: Mitigated via strict JSON schema enforcement and regex JSON sanitizers.

Ambient Noise & Broken Transcripts: Mitigated via dual-phase text cleaning before passing to the analysis engine.

Ambiguous Deadlines & Owners: Mitigated by human-in-the-loop Task Approval Modal before task finalization.

Future Advancements & Product Roadmap

1. Speaker Diarization

Integrate PyAnnote / WhisperX to automatically identify and isolate individual speakers with exact audio timestamps.

2. PM Tool Integrations

Direct webhook syncing of approved action items to Jira, Asana, Trello, Notion, and Google Calendar.

3. Meeting Bot Integrations

Autonomous meeting bot that joins Zoom, Google Meet, and MS Teams calls to record and analyze automatically.

4. Conversational RAG Archive

Chat with historical meetings: Ask natural language questions across months of meeting transcripts.

CONCLUSION & SUMMARY

MeetingLens: Execution Without Friction

- Eliminates post-meeting documentation fatigue through automated intelligence.
- Enforces team accountability with human-in-the-loop task approval.
- Provides a single source of truth for organization-wide meeting decisions.

Thank You! Questions & Discussion Welcome.